GigShield 🛡️

AI-Powered Parametric Income Protection for India's Food Delivery Workers

Guidewire DEVTrails 2026 — Phase 1 Submission

The Problem

India has **15+ lakh active food delivery workers** on Zomato and Swiggy alone, earning roughly ₹27,700/month on average. But when a red-alert flood hits, that income drops to zero instantly. No sick leave. No rainy-day fund. Just a rider sitting at home watching rent money disappear.

Health and accident insurance already exist for these workers — Zomato and Swiggy provide basic coverage, ACKO offers micro-health plans, and PM-JAY is being extended to gig workers under Budget 2025. The gap nobody has filled: **income protection against external disruptions** — weather, curfews, app outages — triggered automatically and paid out instantly via UPI.

That is GigShield.

Why Not E-Commerce or Grocery Delivery?

We evaluated all three personas before locking in food delivery:

Persona	Disruption Frequency	Income Drop Measurability	Fraud Signal Clarity
Food Delivery (Zomato/Swiggy)	Highest — daily weather impact	Direct and immediate	Strongest — order data is real-time
Grocery/Q-Commerce (Zepto/Blinkit)	Medium — warehouse-based, less weather exposed	Moderate	Medium
E-Commerce (Amazon/Flipkart)	Lowest — hub-and-spoke model is more weather-resilient	Harder to isolate	Weakest

Food delivery workers are most exposed, most measurable, and most underserved. That is why we chose them.

Persona — Ramesh, 26, Mumbai

- Earns ₹500–₹900/day on good days, thinks in weekly cycles
- Works 10–14 hour shifts on a ₹6,000–₹10,000 Android, pays via UPI
- Lost ₹8,000–₹12,000 last monsoon season with zero compensation
- Distrusts insurance because "claim process was too complicated last time"
- Does not have time for a 10-step form. Needs decisions in seconds, not days.

Competitor Landscape — Where GigShield Sits

Product	Provider	What It Covers	What It Does NOT Cover
Delivery Partner Insurance	Zomato / Swiggy	Accident, death during delivery	❌ Income loss due to weather or outages
Sachet Health Plans	ACKO, Digit	Short-term hospitalization	❌ Income loss
PM-JAY Extension (2025)	Government	Health coverage up to ₹5 lakh	❌ Income loss
Crop Parametric Insurance	PMFBY, SEWA	Farmer income from weather	❌ Urban gig workers
GigShield	**Our Platform**	**Income loss from weather, curfews, app outages**	❌ Health, accident, vehicle (by design)

No product in the market today covers income loss for urban gig delivery workers on a weekly model triggered by external disruptions. That is our exact white space.

Weekly Premium Model — The Financial Logic

Monthly premiums feel like a large commitment to someone earning daily. Weekly pricing aligns with how delivery workers already think — "kitna hua is hafte."

Tier	Weekly Deliveries	Weekly Premium	Max Weekly Payout
Starter	Less than 50	₹29	₹800
Regular	50–100	₹49	₹1,400
Pro	100+	₹69	₹2,000

How we arrived at these numbers:

A Regular tier worker averages 75 deliveries/week at roughly ₹40–₹50 per delivery — approximately ₹600/day or ₹4,200/week income. The ₹49 premium represents roughly **1.2% of weekly income**, which is consistent with micro-insurance affordability benchmarks across Southeast Asian gig worker studies.

The max payout of ₹1,400 represents roughly **33% income replacement** for a disrupted week — calibrated to be meaningful without creating a perverse incentive to fake claims (full replacement would incentivize fraud).

Week 1 is always ₹29 — the Trust Week. The rider experiences a real payout before committing to higher premiums. Micro-insurance trials in Southeast Asia show trial-first models improve 30-day retention by 40–60%.

Loss Ratio Acknowledgment: Four consecutive red-alert weeks (which happened in Mumbai in July 2024) would break a standalone premium model. Our financial architecture requires a **reinsurance layer** — structurally identical to how Nagaland's parametric agriculture scheme partnered with Munich Re and Swiss Re to absorb catastrophic seasonal risk. Our licensed IRDAI-partner insurer manages this relationship directly.

Parametric Triggers — What Fires a Payout

Every trigger requires **two independent source confirmations**. Single-source parametric insurance failed catastrophically in Nagaland (2021–2024) — NASA satellite data diverged so far from ground reality that zero payouts triggered across three years despite continuous premium collection. We are not repeating that design error.

Trigger	Primary Source	Confirmation Source	Payout/hr	Real Precedent
Heavy Rain above 50mm/hr	IMD API	AccuWeather hyperlocal	₹200	IMD red alerts, Mumbai 2024
IMD Red Alert issued	IMD official feed	Geo-tagged news API	₹300	Documented 4-week Mumbai event
Extreme Heat above 45°C	OpenWeatherMap	IMD heat wave bulletin	₹150	Delhi heat wave 2024
Flash Flood	IMD + NDMA alert	Local traffic API	₹350	Chennai floods 2023
Curfew or Strike	Govt notification API	Local news NLP	₹250	Manipur 2023, various city bandhs
App Outage above 2 hrs	Downdetector public API	Platform status page	₹100	Zomato/Swiggy AWS outage, April 2022 and March 2024

On the App Outage Trigger: This is not speculative. Downdetector publicly tracks Zomato and Swiggy outages in real time and has documented multiple nationwide outages — including a simultaneous AWS-caused Zomato and Swiggy outage in April 2022 that affected thousands of riders for nearly an hour, and a Swiggy outage in March 2024 that

peaked at 2,500+ user reports. Downdetector's data is publicly accessible and API-queryable. No private platform agreement is required.

AI and ML Architecture

1. Zone-Based Dynamic Risk Scoring

We build a risk profile for every pin code using 5 years of IMD rainfall data, NDMA flood zone classifications, and seasonal disruption frequency patterns. A worker in Dharavi pays slightly more than one in Andheri West because the flood history genuinely differs. The model recalibrates weekly and shows the worker exactly why their premium is what it is — transparency builds trust.

2. Personal Behaviour Baseline (Weeks 1–3)

During onboarding, we silently build a behaviour fingerprint per worker:

- Typical active hours and shift patterns
- Average GPS movement per hour
- Regular WiFi hotspots — restaurants, dark kitchens, home network
- Normal battery drain during active versus resting periods

This baseline drives fraud detection — not a rule list, but a unique living pattern per rider.

3. Three-Light Claim System

- 🟢 **Green (~90% of claims):** Device signals match disruption physics. Payout in under 60 seconds. Rider does nothing.
- 🟡 **Yellow (~9%):** Signals inconsistent. One warm request sent: *"Quick selfie and say 'stuck at Bandra' to confirm."* Google Vision checks rain/flood context. Whisper AI verifies voice matches location. Approved in 90 seconds.

- 🔴 **Red (~1%):** High fraud score. **70% provisional payout goes out immediately.** Rider never waits with nothing. Full review in 5 minutes.

Adversarial Defense and Anti-Spoofing Strategy

Directly addressing the Phase 1 scenario: 500-rider syndicate using GPS-spoofing apps to drain the liquidity pool.

1. Differentiation — Real Rider vs. Bad Actor

GPS coordinates alone are useless for fraud detection. A spoofed coordinate is mathematically identical to a real one. Our approach is **Chaos Fingerprinting** — reading the physics of the environment, not just the reported location.

A genuine rider in a monsoon creates:

- Accelerometer micro-vibrations from wind and rain on the phone body
- Degraded GPS signal quality — heavy rain disrupts satellite lock
- Elevated battery drain from weak signal and constant tower searching
- Rapid network tower switching from unstable flood-zone connectivity
- Unfamiliar WiFi SSIDs — not the regular home network

A spoofer at home creates a suspiciously clean signal:

- Near-zero accelerometer movement
- Normal battery drain on home WiFi
- Connected to the same home SSID as every non-working day
- Locked to one stable cell tower

Published research confirms IMU-based GPS spoofing detection is statistically valid — comparing GPS-reported acceleration against actual accelerometer readings creates a

measurable inconsistency spoofing apps cannot hide without additional hardware.

Digital Exhaust Fallback (for ~70% of riders on budget phones with limited sensors): Every Android passively leaves data trails requiring no special hardware:

- Home WiFi SSID active during a claimed "stranded outdoors" scenario → strong fraud signal
- Zero Zomato/Swiggy app activity during claimed disruption hours
- Battery drain pattern inconsistent with outdoor active usage
- Claimed location inconsistent with established delivery route history

Faking a spoofed GPS coordinate takes one app. Faking every single Digital Exhaust signal simultaneously is exponentially harder — it requires the bad actor to also simulate their phone behaving as if outdoors in a storm while sitting at home.

2. Data — Detecting a Coordinated Ring

A single fake claim is hard to catch. Five hundred coordinated ones leave statistical traces:

- **Temporal spike detection:** 50+ claims from the same zone within 10 minutes. Real disruptions build gradually. Coordinated fraud spikes instantly and simultaneously.
- **Network proximity clustering:** Multiple claimants recently connected to overlapping WiFi networks — physical co-location despite claiming geographic spread.
- **Onboarding graph analysis:** Accounts registered together, mutual referral chains, shared device fingerprints — signs of organized recruitment without monitoring any external communication platform.
- **Environmental consistency check:** During a genuine red alert, all affected workers show similar Digital Exhaust patterns. A subset showing suspiciously clean, stable signals while claiming disruption gets collectively flagged for ring-level review.

3. UX Balance — Protecting Honest Workers

False positives are a greater operational risk than fraud. A genuine worker with a network dropout in a flood looks suspicious by definition.

Our principles:

- **Never hold a full payout.** Flagged claims get 70% immediately. Remaining 30% releases within 5 minutes.
- **Yellow Light feels like help, not interrogation.** Warm tone, one simple action, 90-second resolution.
- **2-hour network drop grace period.** Going offline during a declared event is expected. We cover it automatically.
- **Repeated false flags trigger human review**, not automatic rejection. Written explanation within 24 hours either way.

Business Model and Legal Structure

GigShield is a **B2B2C technology platform**, not a licensed insurer. We partner with an IRDAI-licensed company — ACKO, Digit Insurance, or SBI General — who underwrites the policies. GigShield provides the risk engine, trigger monitoring, fraud detection layer, and the complete worker-facing product experience.

This is the only viable path under Indian insurance regulation. IRDAI's "Insurance for All by 2047" initiative and the incoming Bima Vistaar scheme signal strong regulatory support for this exact product category. Licensed insurers have been actively seeking technology partners to reach the uninsured gig segment — we are building the technology layer they need.

Tech Stack — And Why

Layer	Technology	Justification
Frontend	React / Next.js (PWA)	Mobile-first, works from a WhatsApp link, no Play Store friction
Backend	Java / Spring Boot	Production-grade reliability for policy management, claims orchestration, and real-time API handling. Our core developer's primary stack — we ship what we know deeply, not what looks impressive on paper.
ML Microservice	Python / FastAPI	Our ML developer's primary environment. Scikit-learn for risk scoring, Isolation Forest for anomaly detection. Deployed as an independent service — independently scalable, independently testable, cleanly separated from business logic.
Database	PostgreSQL + Redis	PostgreSQL for policy and claims data. Redis for real-time trigger state and session caching.
Weather APIs	IMD + OpenWeatherMap + AccuWeather	Multi-source confirmation — no single point of failure
Outage Detection	Downdetector public data + platform status pages	Publicly documented and API-queryable. No private agreements needed.
Payments	Razorpay UPI (test mode)	Industry standard in India, instant UPI payout simulation

Why Java + Python microservice instead of a unified stack:

Combining Java and Python is not a compromise — it is the correct architecture for this problem. The core platform (policy creation, premium calculation, claims orchestration, user management) needs the stability, type safety, and transactional reliability that Spring Boot provides. The fraud detection and risk scoring models need the ML ecosystem — scikit-learn, pandas, numpy — that Python provides natively. Running them as separate services means each can be deployed, scaled, and updated independently. A bug in the ML model does not take down the claims system. This is how production insurance systems are actually built.

Why PWA over native app:

Our user should not have to find us on the Play Store before experiencing value. A PWA works directly from a WhatsApp link shared in a Zomato or Swiggy delivery community group. Full onboarding in under 3 minutes. For a demographic where app install fatigue is real and storage space on budget phones is limited, this friction difference is enormous.

Risks — We Are Not Hiding These

Risk	Severity	Mitigation
IMD data latency 3–4 hrs	High	Multi-source confirmation + hyperlocal API cross-check
Monsoon season loss ratio spike	High	Reinsurance layer via licensed insurance partner (Munich Re / Swiss Re model)
Budget phones with no IMU sensors	Medium	Digital Exhaust fallback covers this segment entirely
IRDAI licensing requirement	Medium	B2B2C model — we are the technology layer, partner insurer holds the license

Risk	Severity	Mitigation
Worker trust deficit in insurance	Medium	Trust Week at ₹29, instant UPI payout, transparent zone risk UI
Advanced IMU spoofing by attackers	Low	Digital Exhaust layer adds independent signals that cannot all be faked simultaneously

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Development Roadmap

- **Phase 1 (Mar 4–20):** Research, architecture design, README, 2-min concept video ✅
- **Phase 2 (Mar 21–Apr 4):** Working MVP — registration, policy creation, dynamic premium engine, 3–5 automated parametric triggers, GPS + Digital Exhaust fraud detection baseline
- **Phase 3 (Apr 5–17):** Full Chaos Fingerprinting, coordinated ring detection, 70/30 provisional payout system, dual analytics dashboard (worker + insurer view), Razorpay UPI payout simulation, final pitch deck

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GigShield — because 15 lakh riders deserve a safety net that actually works when the rain hits.

Repository: [GitHub Link — add before submission] **Demo Video:** [2-Minute Video Link — add before submission] **Team:** [Team Name] | Guidewire DEVTrails 2026